Disaster-Proofing North Stradbroke Island with Community Sovereign Kiosks

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Introduction: Resilience Challenges and Vision

North Stradbroke Island (Minjerribah) faces escalating climate and systemic risks – from cyclones and bushfires to the prospect of civilisation-level disruptions (e.g. a solar super-flare or a coordinated cyber attack). In such scenarios, mainland power, communications, food supply, and transport links (ferries) could be severed for extended periods. The island's disaster plan must therefore assume periods of total isolation and grid-down conditions. Drawing on the island's innovative "Minjerribah Integrated Systems Vision" , this plan proposes a network of offline-first, low-energy, solar-powered community kiosks. These kiosks (each no larger than an ATM) will function as "Sovereign Nodes" – independent, resilient hubs that keep critical services running locally even when outside systems fail. By leveraging open-source technology and community ownership, the kiosk network will fortify Minjerribah against natural disasters and even existential events, all while empowering the Quandamooka people's data sovereignty and self-reliance.

Goal: Establish a Community Sovereign Kiosk Network that can operate island-wide without external power or internet. Each kiosk will provide multiple community functions – local data storage, emergency communications, democratic participation, coordination of food and energy resources, and access to the island's "digital twin" (Virtual Minjerribah) – supported by robust hardware that withstands cyclones and bushfires. The following sections detail the kiosk design, capabilities, integration with existing initiatives, a milestone-based rollout strategy, and matching funding pathways (with an emphasis on 2025–2026 grant opportunities) to turn this vision into reality.

Kiosk Design for Extreme Resilience

Each kiosk is a standalone solar-powered unit engineered for harsh conditions. It will be housed in a hardened enclosure (steel or composite) anchored securely above flood lines and designed to survive Category D cyclone winds and intense bushfire heat. For example,

industrial off-grid “power kiosk” designs already exist that are “resilient to Australia’s extreme and harsh weather conditions” (wind-rated for cyclones) and can be “deployed…following a major disaster, such as a bush fire” . The Straddie kiosks will similarly incorporate:

- Solar PV panels and battery backup: Sufficient solar generation and deep-cycle battery storage to run all electronics for several days of cloudy weather. No external fuel is needed, eliminating reliance on diesel. Panels will be securely mounted (or even retractable) to prevent cyclone damage.

- Low-power, EMP-resistant electronics: The computing core will use efficient processors (e.g. ARM-based single-board computers) and minimal moving parts. Critical components can be housed in Faraday-shielded casings to improve survivability against electromagnetic pulses (as might occur in a solar Carrington event). Spare critical parts could be stored in EMP-proof containers on-island for backup.

- Rugged communication gear: Each kiosk will have built-in long-range radio communication (such as a LoRa mesh radio, UHF/VHF emergency transceiver, or satellite link) to connect with other kiosks or external responders during outages. Antennas will be reinforced or retractable. In normal conditions, kiosks act as Wi-Fi hotspots and cellular signal boosters; in emergencies they switch to independent radio networks.

- Durability against heat and fire: Enclosures will use fire-resistant materials (metal or specialized composites) and internal cooling systems. They can be located in cleared, defensible spaces or co-located with community fire shelters. If a bushfire threatens, the kiosk can shut down non-essential systems to avoid overheating, while the battery and core electronics remain protected in insulated compartments.

Design Summary: These kiosks are essentially micro off-grid facilities that remain operational through extreme weather. They are compact (ATM-sized) yet built to infrastructure-grade standards, akin to hardened emergency communication cabinets or power kiosks. Community input (especially Indigenous knowledge of local conditions) will guide site selection – e.g. placing kiosks on high ground safe from storm surge, away from dense fuel loads, and at central gathering points in each settlement.

Offline-First Technology Stack

A cornerstone of this plan is an offline-first, peer-to-peer software architecture that allows kiosks to function autonomously and sync with each other when possible, without relying on any central server. This approach aligns with the Virtual Minjerribah project’s “Sovereign Node” paradigm, in which each node holds a complete local copy of data and the network achieves consensus via math rather than authority. Key technical principles include:

- Conflict-Free Replicated Data Types (CRDTs): All critical data (messages, sensor readings, voting records, etc.) will be stored in CRDT structures. CRDTs enable optimistic replication – kiosks can accept updates locally even while offline, and

whenever they reconnect (via mesh network, physical data swap, or intermittent internet), their states merge without conflicts into a consistent whole. This ensures that after an island-wide outage, each kiosk can reconcile its offline transactions (community votes, reports, edits) with others seamlessly.

- Peer-to-Peer Networking & Open Protocols: Kiosks communicate over local radio links or any available channels using open standards (e.g. Wi-Fi Direct, LoRaWAN, Bluetooth mesh, and IP-based delay-tolerant networking). No proprietary or cloud-dependent systems are required. For example, if two kiosks come within range (or a portable device ferries data between them), they automatically sync datasets via P2P protocols. Standard file formats and open APIs allow integration with external systems when connections restore.

- Decentralized Identity and Security: To operate offline and still maintain trust, the kiosks use Decentralized Identifiers (DIDs) and cryptographic signatures for data. Users and local officials will have digital IDs allowing them to sign votes or emergency instructions, which kiosks can later verify when syncing. This prevents tampering and ensures authenticity without needing a central authority. Even governance of the network itself can be community-driven – each kiosk is effectively owned by the community and participates in a “distributed governance” model of data sharing.

- Minimal bandwidth usage: Recognizing that post-disaster links will be constrained, the system prioritizes text and sensor data over heavy media. Kiosks maintain local content caches (e.g. storing key maps, manuals, and digital library files) so that residents can access information without internet. When synchronization occurs, it sends only compact diffs or aggregated data, often on low-bandwidth channels. This “delay-tolerant” design ensures useful exchange even via short radio contacts or high-latency satellite bursts.

Result: The kiosk network behaves like a resilient local internet that can fall back to an information “sneakernet” if needed. Even in a worst-case scenario of island-wide isolation, each kiosk is a self-sufficient server that keeps vital apps running. When connectivity is restored (whether through mobile networks, Starlink satellite, or a boat carrying a hard drive), the kiosks exchange their CRDT updates to become consistent. This architecture was chosen because, as the Virtual Minjerribah team noted, “traditional client-server models” would create a single point of failure and “cannot fulfill the mandate for sovereign nodes”, whereas a CRDT-based P2P network “makes true offline-first functionality possible” for a shared system.

Multi-Function Community Kiosks

Each kiosk will be multi-purpose, serving day-to-day community needs and critical functions during crises. By combining roles, the kiosks maximize value and justify ongoing community investment. The core capabilities include:

- 1. Local Data Vault and Knowledge Hub: Every kiosk stores important databases and documents locally for access anytime. This includes cached internet content (weather forecasts, news feeds, maps), emergency manuals, and cultural knowledge archives. For Minjerribah, the kiosks can host Indigenous knowledge recordings and the Virtual Minjerribah digital twin data so that users can explore the island's past and future scenarios without internet . All cultural data respects Indigenous Data Sovereignty protocols – sensitive information is protected by permissions governed through Quandamooka protocols (as outlined in the Virtual Minjerribah plan) . In everyday use, community members might visit a kiosk to look up local history, browse a digital library, or contribute local observations (e.g. environmental readings, oral histories) which are stored on-island and later synced to the wider digital twin when connectivity returns.

- 2. Emergency Communication Node: In a disaster, the kiosks become life-saving communication points. They will automatically switch to emergency mode, broadcasting local alerts and instructions (via siren, speaker, or LED display) even if phone networks are down. Residents can come to the kiosk to send text messages or emails off-island through any remaining channel (for example, a kiosk could route a short SMS via HF radio or connect to an overhead emergency drone network). Likewise, rescue authorities can drop information packets (via satellite or radio) to kiosks, which will relay the messages to the local populace (through on-screen bulletins or local Wi-Fi). Each kiosk doubles as a community noticeboard – in offline conditions, it runs a BBS-style messaging system where people can post needs or offers (medical help, spare fuel, etc.) that will sync with other kiosks when possible. By design, no single point of failure exists: even if only one kiosk has a satellite uplink, it can distribute incoming messages to others over the island's mesh. This aligns with federal efforts to improve disaster communications – for instance, the STAND program has installed satellite Wi-Fi at over 1,000 evacuation centers and is expanding to "500 additional emergency sites" so that communities stay connected when mains networks go down . The kiosks will leverage such capabilities, effectively acting as mini evacuation communication hubs in each neighborhood.

- 3. Community Voting and Civic Services: The kiosks support offline e-governance and community decision-making, a concept in line with the island's push for "gamified democracy" and stronger local voice . They will run a secure voting application where residents can cast votes on community decisions or local elections using their digital ID. Votes are stored locally (as signed records) and merged with other kiosks via CRDT sync, ensuring an accurate tally without a central server. Similarly, the kiosks can host petitions, surveys, and a transparent "community ledger" of decisions. This provides a resilient form of governance during disasters – for example, if cut off from mainland authorities, the island's emergency committee can poll residents on resource allocation or recovery plans via kiosks. In non-emergency times, this platform encourages ongoing civic engagement (townhall polls, feedback on council proposals, etc.), bridging the digital divide for residents who may not have internet at home. Data privacy is paramount: personal inputs stay at the L1 community level and only anonymized aggregates are shared upwards (consistent with the "Fractal Governance" model that "enforces privacy" by design). Each kiosk essentially acts

as a trusted local “ballot box” and civic terminal owned by the community.

- 4. Emergency Coordination & Resource Sharing: During crises, the kiosks transition into coordination centers for disaster response. They will run an incident management interface accessible to trained volunteers and first responders. Community emergency response teams (including QYAC rangers, SES, etc.) can log updates at the kiosk – e.g. road blockages, fire outbreaks, injured persons, or supply status. This information is stored on the kiosk and shared with others over the mesh, creating a real-time local situation report even if external networks are down. The kiosks will also facilitate resource coordination: an integrated bulletin board or inventory system lets residents and local farms record available supplies (water tanks, generators, produce, medical supplies) and needs. For instance, if ferries stop and food imports cease, kiosks can display a live community larder – showing which co-op or household has surplus vegetables or fishing catch, and who needs food – enabling direct bartering or distribution. This ties into the island’s “Full Spectrum Consumption” and local food security initiatives (simulated in the digital twin) . The kiosk essentially becomes a local supply chain node, coordinating “island-wide rollout of Garden Mate hydroponics” or similar projects by tracking production and demand . In emergency mode, it prioritizes essentials: e.g. a resident could submit “need baby formula” at one kiosk, and another community member with extra formula could see that request on their nearest kiosk and respond. All such exchanges can be logged (potentially even rewarding contributors with Community Hours (C-Hours) under the island’s Braided Economy model).

- 5. Digital Twin Access Point: A remarkable feature is providing “Virtual Minjerribah” access at each kiosk. Virtual Minjerribah is the island’s evolving 4D digital twin and simulation sandbox . The kiosks will have interfaces (e.g. a rugged touch-screen or the ability to stream to personal devices over Wi-Fi) for community members to engage with these simulations. In practical terms, this means even without internet, locals can view critical simulations: for example, a bushfire spread model for their area, or a flood map showing safe zones if a cyclone-driven surge occurs. Because each kiosk runs as a sovereign node of the simulation, it can store and update a portion of the digital twin data. The architecture using CRDTs ensures that when a connection is available, the kiosk pulls in any global updates (e.g. latest weather or sensor data) and also shares local inputs (like community sensor readings or anecdotal reports) back to the global model . This empowers residents and local decision-makers: they can “test policy decisions before real-world implementation” using the kiosk’s simulation mode, even if the mainland is offline . For instance, an island councilor at the kiosk could adjust a virtual scenario (like closing a certain road during an evacuation) and see the simulated impact on traffic flows in real time. In essence, the kiosks act as portals to a local copy of the island’s digital twin, supporting planning and education. Grounding the high-tech digital twin in a physical kiosk ensures inclusivity – everyone, including those without personal high-end devices, can benefit from these insights by simply visiting the kiosk.

- 6. Microgrid Control Station: As the island moves toward renewable energy independence (via projects like Straddie Clean Energy Co.), the kiosks will help manage and democratize the community microgrid. Each kiosk connects to local

energy assets – solar farms, battery banks, community biogas, etc. – through open standard protocols (e.g. Modbus, IEEE 2030.5) provided by the microgrid operator. In daily use, a kiosk can display the local grid status: current power production, storage levels, and usage in that area. Community members could check energy prices or availability and make informed choices (for example, seeing that batteries are low and choosing to conserve power). Crucially, in a grid-down scenario, the kiosks coordinate islanded operation. They can execute load-sharing algorithms and communicate with smart inverters to balance the microgrid. If part of the island loses generation, kiosks in that zone can orchestrate a controlled shed of non-critical loads while keeping critical facilities powered (like clinics or water pumps). This peer-to-peer grid management echoes the regenerative system approach where "sand, sun, and sea" (the island's resources) are harnessed for "energy independence" . Kiosks may also allow residents to participate in energy decisions – e.g. voting on whether to allocate reserve power to the desalination plant or to refrigeration during a long outage. By providing a local interface to the microgrid, the kiosk ensures energy resilience is in the hands of the community. Technically, even if the central SCADA system is down, the kiosks and inverters can form a distributed control loop to keep the lights on in island micro-sections. This concept will be tested via the Virtual Minjerribah simulator – e.g. stress-testing the grid against a "Carrington-level geomagnetic storm" scenario to validate that kiosk-mediated controls maintain stability .

In summary, each kiosk is a multi-tool for community resilience. On a normal day, it might be a digital café (solar-powered device charging and Wi-Fi), a cultural showcase, or a training portal. When shocks hit, it transforms into an emergency lifeline, coordinating vital needs. The combination of everyday utility and disaster functionality ensures the kiosks remain well-maintained and familiar to locals, rather than just dusty emergency infrastructure. This approach mirrors the successful model of hubs like the Kowanyama Community Hub in Far North QLD, which is funded as a "multipurpose Disaster Coordination Centre" but in non-disaster times serves as an Indigenous Knowledge Centre and community program site . Straddie's kiosks will be woven into community life in the same way, so that resilience is built and exercised year-round.

Integration with Local Initiatives and Ecosystem

This plan does not exist in isolation – it deliberately braids into Minjerribah's ongoing innovation ecosystem and Indigenous governance frameworks. Key integrations include:

- Quandamooka Sovereignty and Data Governance: The kiosks embody the principle of Sovereign Data. All cultural and community data collected/stored locally remains under Quandamooka control per Indigenous data sovereignty protocols . The governance layer for who can access what in the kiosk (especially for cultural knowledge or personal data) will be co-designed with QYAC (Quandamooka Yoolooburrabee Aboriginal Corp). In practice, this could mean implementing role-based access using DIDs – e.g. certain sacred knowledge modules on the kiosk require a Quandamooka elder's digital key to unlock. This ensures technology

strengthens rather than undermines Traditional Owner custodianship of knowledge. The Sovereign Node approach, as espoused in Virtual Minjerribah, is fundamentally about aligning digital sovereignty with real-world sovereignty . By making each kiosk a fully functional node that “inverts the power structure” of centralized systems , we ensure the island’s digital infrastructure reflects self-determination. Community ownership of the kiosks (through a co-op or trust) is critical – they will be assets owned and managed by island residents (with QYAC as a key stakeholder), not by external corporations.

- Virtual Minjerribah & Digital Twin: The kiosks are effectively the physical extension of Virtual Minjerribah on the island. Virtual Minjerribah is the comprehensive digital twin project mapping the island’s past, present, and future scenarios . Each kiosk will run a node of this system, meaning it stores a portion of simulation data and allows users to interact with it offline. This federated approach follows Virtual Minjerribah’s architecture where every user device is a node of the simulation . The kiosks will sync with the main Virtual Minjerribah database whenever possible, exchanging updates using the CRDT mechanism described earlier. By doing so, the island’s real-time data environment stays alive locally during outages – sensors feeding into the twin (like weather stations, tide gauges, or wildlife trackers) can log data to the nearest kiosk, which later merges into the island-wide model. Conversely, globally-run simulations (e.g. a tsunami model from the Global Sensorium network) can be downloaded to kiosks when connectivity permits, so the community can benefit from broader intelligence. The kiosks also serve as public front-ends for Virtual Minjerribah’s community engagement features: gamified planning workshops, ancestor stories in AR, or “what-if” scenario games can be hosted at kiosks during community events to involve those without personal VR equipment. This integration ensures the digital twin is accessible to all and remains operational under all conditions, greatly enhancing both disaster preparedness and cultural education.

- Aura of Intelligence (Health and Aged Care): The Aura of Intelligence initiative is a cutting-edge, AI-driven health platform being developed to “provide groundbreaking support for dementia and aged care” . Integrating Aura with the kiosk network offers two benefits. First, each kiosk can host an “Aura Capsule” – essentially a local instance of the AI health assistant that can operate offline. This would allow elders or caregivers to use cognitive tools, memory aids, or health monitoring at the kiosk even if the internet is down. For example, an elder could have their “digital companion” (part of AuraOS) available at the kiosk to help retrieve important reminders or exercises for brain health. Second, in emergencies, the kiosk can store medical data and facilitate telehealth: since Aura aims to optimize human health, an offline mode could include first aid advice, a repository of patient health records (with consent), and even AI-driven triage support. If a senior with dementia is lost or confused during a crisis, they could approach a kiosk which recognizes them (via secure biometric or DID) and communicates calming guidance or alerts caregivers. The clinical data security of Aura fits well with the kiosk’s sovereign data approach – sensitive health metrics would stay encrypted on the kiosk and only sync to clinical researchers under agreed protocols. By demonstrating Aura’s usefulness in connectivity-challenged settings, we also strengthen its case for funding (e.g. state and federal health grants) . In short, the kiosk network provides Aura with a real-world deployment platform on

the island, extending its reach to those without smartphones and ensuring continuity of care in disasters. This partnership would be an early win for the broader ecosystem, showcasing Minjerribah as a testbed for “AI-driven community healthcare in a resilient, offline-first context.”

- SETCo (Straddie Employment & Training Company): SETCo – described as the island’s “central human capital engine” with a mandate to “provide training, onboarding, job pathways for locals” – is integral to this plan. The deployment and maintenance of the kiosks will create new skilled roles (tech maintenance, solar installers, data managers), and SETCo will ensure locals are trained to fill them. We will coordinate with SETCo’s “Ready SET Co.” curriculum to incorporate modules on kiosk hardware servicing, networking, and data governance. This aligns with SETCo’s model of issuing Verifiable Credentials to trainees’ digital wallets upon skill completion – for example, a local youth might earn a “Resilient Systems Technician” credential after learning to operate the kiosk. Those credentials are perfectly in step with our system (the kiosk can even host the Sovereign Skills Wallet data for local verification) . Beyond training, SETCo will use the kiosks as outreach and service points. They can push job notices, host interactive learning content, or facilitate “tele-training” sessions through the kiosks for those who cannot travel. In a disaster scenario, SETCo-trained teams on-island will be the ones to respond first; thanks to kiosk coordination, they can organize effective volunteer actions (tree clearing, distributing supplies, etc.) and log their contributed hours. These hours may translate into C-Hour community credits, strengthening the Braided Economy feedback loop where community work (like disaster recovery labor) is recognized and rewarded . Essentially, the kiosk network will not only be built by local hands (with SETCo’s help) but will continue to upskill the community – a tangible example of technology creating resilient local jobs.

- Straddie “Sovereign Node” Ecosystem: The kiosks reinforce the broader notion of Minjerribah as a Sovereign Node in global systems. By design, the island’s infrastructure should be able to detach and operate independently. The kiosks anchor this capability in practice. For instance, if a global “black swan” event occurs (financial collapse, internet blackout), Minjerribah’s key governance and commerce systems can fail gracefully over to the kiosk network. Trade can continue internally via the Reciprocity Ledger and C-Hour currency logged on kiosks , and decisions can still be made via local voting. This makes the island a pilot site for the “Civilization of Sand” concept – a closed-loop, regenerative economy that could persist even if broader civilization stumbles . As noted in strategy documents, “humanity faces converging existential risks” and needs “a new, decentralized paradigm” for resilience . Minjerribah’s sovereign kiosk network could serve as a model for decentralized resilience – potentially linking with a global federation of other community nodes in the future. In Year 2 of the roadmap, for example, Virtual Minjerribah envisions connecting to a “Planetary Digital Twin (Global Sensorium)” . Our kiosks would enable Minjerribah to contribute local data to this global effort and pull down insights (like the Space Weather Hub’s solar storm alerts) for local use . The sovereignty stack – from DIDs to distributed ledgers – is thus instantiated on the ground through these kiosks. They are where high-level concepts meet practical implementation, ensuring that all the grand ideas (digital twin, braided economy, etc.) have a

community touchpoint accessible to every island resident.

By tightly integrating with these initiatives, the kiosk plan amplifies their impact and vice versa. It ensures cohesion in the overall Minjerribah Innovation Ecosystem: energy, food, culture, governance, and technology all converge at the kiosk. This integration will be emphasized in grant applications and partnerships – showing funders that the kiosks aren't standalone widgets but part of a "deeply interconnected system" driving a resilient, inclusive economy . The result is a holistic strategy where technology reinforces community and culture, rather than replacing them.

Milestone-Based Implementation Strategy

Rather than rigid timelines, this roadmap is organized into flexible milestones that mark capability targets and community readiness. Each milestone builds on the last, allowing the project to adapt to funding availability and evolving needs:

- Milestone 1: Community Co-Design & Pilot Kiosk – "Foundation and Partnership". This first stage focuses on co-designing the kiosks with the community and establishing governance. Actions: Formalize the partnership with QYAC and local stakeholders (as recommended, a "sovereign partnership with the Quandamooka Nation" is the non-negotiable first step) . Host design workshops with residents, elders, and youth to decide kiosk locations, aesthetic design (incorporating Quandamooka art and language), and priority functions. Concurrently, secure seed funding for a prototype (via grants or philanthropy – see Funding section). Outcome: Deploy a Pilot Kiosk at a central location (e.g. Dunwich Community Hall or similar safe site). This unit will be a minimally functional prototype: solar power, local server running basic apps (messaging, data storage), and a simple interface. It serves as a proof of concept and a community demo unit. Success Criteria: Community feedback gathered, QYAC governance protocols integrated (e.g. a community data license established), and at least one initial kiosk operational by the end of this phase.

- Milestone 2: Core Network & Resilience Features – "Build the Essentials". Using lessons from the pilot, scale up to a basic network of kiosks across the island's key settlements (Dunwich, Amity Point, Point Lookout, and potentially smaller communities/camps). Actions: Construct additional kiosks (perhaps 3–5 units) with improved hardware (robust enclosures, higher solar capacity) and link them via a long-range mesh (e.g. LoRa radios or point-to-point Wi-Fi links between towns). Implement the core offline-first software stack on all units: this includes the CRDT database, digital ID system, and a minimal suite of applications (offline chat board, file library, basic voting tool). Work with local emergency services to integrate the kiosks into disaster response plans – e.g. program automated weather alerts, ensure each SES depot has a way to interface with kiosk data. Start testing extreme scenarios: conduct a "island disconnect drill" where internet and mains power are shut off to simulate an outage, and evaluate how kiosks perform handling communications and power sharing. Outcome: A resilient kiosk mesh network covering the island's population centers, demonstrated to work for at least 72 hours

in fully off-grid mode. Key resilience features (solar power, battery life, wind/fire durability, basic emergency comms) are validated. Success Criteria: Island-wide coverage achieved (no major community is more than a short walk or drive from a kiosk); successful completion of at least one drill where kiosks managed communications during a simulated outage; and community members are increasingly using kiosks for daily tasks (indicating adoption).

- Milestone 3: Service Expansion & Integration – “Multi-Function Rollout”. With the network backbone in place, this stage adds the rich functionalities described earlier: digital twin access, advanced emergency coordination tools, microgrid integration, and Aura health services. Actions: Deploy the Virtual Minjerribah interface on kiosks – this may involve installing higher-performance computing (GPUs or edge servers) at one or two kiosks to handle 3D simulations, or optimizing the twin for low-spec hardware (potential collaboration with a university or tech partner to create lightweight visualization tools). Integrate the kiosks with the Straddie microgrid: coordinate with the Straddie Clean Energy Co. (or relevant utility) to connect inverters and battery management systems to kiosk controllers via secure APIs. This lets kiosks display energy data and execute demand response commands as needed. Launch the community voting platform on the kiosks, perhaps timed with a local event (e.g. a mock referendum or a QYAC initiative) to test high participation. Importantly, introduce the Aura of Intelligence pilot: outfit one kiosk near the aged-care center or health clinic with Aura’s “Capsule” (sensors and interface for health monitoring) and run a trial where elders use Aura via the kiosk for a few weeks. Throughout this stage, extensive training and outreach is conducted – SETCo will train local kiosk “custodians” and also digital mentors to assist less tech-savvy residents in using the new services. Outcome: The kiosk network evolves into a full-service community grid: people can check the digital twin for planning info, vote or give feedback on local issues, coordinate resource sharing, and access basic healthcare support tools – all through the kiosks. Integration means that data flows between systems: e.g., an IoT soil sensor from a community garden might feed into the Virtual Minjerribah model via a kiosk, or a power outage in one microgrid segment triggers a kiosk to send an island-wide alert. Success Criteria: Multi-functionality proven – at least one real use-case in each domain (governance, energy, food, health) has been delivered. For example, a digital community meeting was held via kiosk where residents at different kiosks participated in a joint planning session (demonstrating the network’s sync capability). Another example: during a heatwave, kiosks successfully coordinated the microgrid to keep critical loads (like medical fridges) running. Such achievements solidify stakeholder confidence.

- Milestone 4: Hardening & Scaling Up – “Resilience Maturity”. This phase focuses on making the kiosk system truly disaster-proof and sustainable long-term. Actions: Incorporate redundancies and upgrades identified in earlier testing. This could mean adding a second solar/battery unit at each site for extra capacity, installing lightning protection and further cyclone proofing, or even burying fiber optic or cable links between kiosks for faster sync when possible. We will also add mobile kiosk units – e.g. a portable version that can be deployed to a pop-up evacuation center or moved to where it’s needed if a certain kiosk is knocked out. At this stage, pursue formal certifications and agreements: get the system recognized in the Redland City Council

disaster management plan, and sign MOUs with state emergency agencies so they know how to interface with it. Run a full-scale simulation of a catastrophe (using the digital twin in concert with physical drills): e.g. simulate a Category 4 cyclone direct hit – pre-run the scenario in Virtual Minjerribah to identify vulnerabilities, then conduct on-the-ground exercises with kiosks handling communications, and refine based on outcomes. Another stream is exploring inter-island or mainland links: perhaps set up a dedicated long-range radio from a Straddie kiosk to a partner node on the mainland (like Redland Council's HQ) as a failsafe channel. Outcome: The kiosk network is battle-tested and fortified. It should withstand protracted crises; even if one or two kiosks fail, others compensate. Maintenance routines are established (with local technicians rotating checks on solar panels, batteries, software updates). The community is well-versed in using the kiosks under stress – regular drills or games (like a "disaster scavenger hunt" via kiosk tasks) keep people familiar. Success Criteria: An external audit or expert assessment is obtained confirming that the system can survive specified stress scenarios (extreme weather, 7+ days off-grid, etc.). Additionally, the island demonstrates self-sufficiency in operating the kiosks – the community-led management (perhaps a Kiosk User Cooperative) is in place, handling everything from content moderation on message boards to technical troubleshooting, without needing permanent off-island assistance.

- Milestone 5: Innovation and Grant Leverage (2026 and beyond) – "Scaling Outward and Continuous Improvement". By now, the kiosks are ingrained in Straddie life and proven in concept. This milestone looks outward to sustain and expand the impact. Actions: Capitalize on the system's success to attract further funding and partnerships. For example, prepare applications for scaling grants or awards (see Funding Opportunities table below) highlighting how the project delivered on government priorities like digital inclusion, smart cities, and resilience. Use the network as a living lab to trial new innovations – perhaps collaborating with CSIRO or universities on next-gen features (AI-driven disaster prediction running on kiosks, or quantum battery interfaces – the sky is the limit). The system should also be prepared for handover and replication: document everything and create toolkits so that other regional communities can adopt the "sovereign kiosk" model. Minjerribah can host knowledge-sharing workshops or publish open guides (maybe through Advance Queensland's platform) to scale this prototype to other islands or remote towns. We will also integrate with any national resilience networks that emerge; for instance, linking the kiosks into the federal Emergency Management system via gateways, or contributing to data repositories. Meanwhile, continue to iterate locally – gather community suggestions for new services (maybe the youth want a gaming LAN party function on kiosks, or fishers want tide charts – these can be added). Outcome: A sustainable, evolving ecosystem around the kiosks. By securing ongoing funding (from cost savings, grants, possibly revenue-generating apps like tourist info services), the kiosks are financially self-sufficient for maintenance and upgrades. The project transitions from a startup phase to an established community infrastructure akin to roads or libraries. Success Criteria: Achieve funding for multi-year operation (for example, a grant that covers maintenance for 5+ years, or community fundraising that contributes to a renewal fund). Also, measure increased resilience quantitatively – e.g. by 2027, emergency response times on the island are improved, fewer service disruptions occur during mainland outages, and digital engagement rates (like % of

residents participating in local e-votes) have risen, demonstrating social resilience. Perhaps by this stage, the Minjerribah model receives recognition (e.g. a Resilient Australia Award) and is being emulated elsewhere – the ultimate validation of its success.

These milestones provide a structured yet flexible path. The approach is adaptive: for instance, if a big cyclone strikes in 2026, we may fast-track certain emergency features (jumping ahead in the plan), whereas if certain funding is delayed, we sustain community engagement at the pilot stage a bit longer. By focusing on capabilities and community readiness at each milestone, we ensure that progress is meaningful even if timelines shift. At every stage, we will evaluate and incorporate feedback, staying true to the core vision: a resilient, community-owned kiosk network that grows organically with the community's trust and skills. This milestone framework also aligns well with funding requirements – many grants will fund specific stages (e.g. a pilot, or a scale-up), so we can match proposals to the milestones.

Funding Strategy and Grant Opportunities

To realize this plan, a diverse funding strategy is crucial, tapping public grants, innovation programs, and partnerships. Fortunately, the project aligns with many current funding priorities (resilience, smart communities, Indigenous innovation, digital inclusion). Below is a table of key 2025–2026 grant opportunities and programs, matched to how they support specific kiosk capabilities or milestones:

Grant / Program	Relevant Kiosk Component	Opportunity Details
Advance Queensland – Ignite Ideas Fund	Prototype Development: Funding to build and test the initial kiosk technology (hardware & software MVP) by a Qld startup.	Grants up to ~$200,000 for innovative QLD-based startups to commercialize new products . Our kiosk project – combining IoT, software, and resilience tech – is an excellent fit. Next round expected in 2026; we will apply to cover Milestone 1–2 costs (design, pilot kiosk).
Advance Queensland – Ignite+ Scale	Scaling & Business Support: As the kiosk network moves from pilot to island-wide	A 2026 scale-up program for high-growth QLD ventures, offering intensive support

	roll-out (Milestone 3–4), this program provides mentoring, networking, and investor exposure.	rather than direct funding . By participating, the project team can refine its business model (e.g. a social enterprise for kiosk tech) and connect with partners. EOI opens late 2025 .
Advance Queensland – Deadly Innovation (Indigenous Innovation)	Indigenous Co-Design & Leadership: Supports involvement of Quandamooka people in tech innovation (the kiosk project led in partnership with QYAC).	The Deadly Innovation initiative backs Indigenous innovators as part of building an “innovation economy” . Programs under this banner (e.g. Deadly Deals grants) can provide funding and support (often $25k–$100k) for projects that empower First Nations communities. Our kiosks, co-designed with Quandamooka and enhancing Indigenous data sovereignty, will be a strong candidate.
Queensland Reconstruction Authority – Disaster Ready Fund (DRF)	Resilient Infrastructure Deployment: Capital funding to manufacture and install kiosks as hardened community infrastructure for disaster management (Milestone 2–3).	The DRF is a federal-state program committing ~$200 million annually (rounds through 2025–26) for resilience projects . QRA coordinates QLD bids; our project fits as a community resilience hub network. Notably, DRF Round 3 funded the $9M “Kowanyama Hub” for disaster coordination and indigenous knowledge – a precedent for our multi-use kiosks. We will target Round 4 (2026) seeking a mid-sized grant (e.g. $1–3M) to deploy the full

		kiosk network and sheltering improvements.
Smart Cities & Suburbs / Thriving Suburbs Program	Digital Community Infrastructure: Funds the kiosk network as a smart community initiative improving liveability, connectivity, and inclusion on Minjerribah.	The Thriving Suburbs Program (2024–27) offers $0.5–15 million for urban/peri-urban community infrastructure, aiming to enhance “amenity, liveability and social cohesion”, with emphasis on resilience and First Nations communities . As an incorporated entity in partnership with Redland City, we could apply for ~$1–2M to cover kiosk construction and digital inclusion outcomes. This federal grant requires 50% co-funding , which we could combine with state funds or in-kind contributions.
CSIRO – Kick-Start Program	R&D and Technical Innovation: Supports partnering with CSIRO experts to solve technical challenges (e.g. optimizing CRDT software, energy systems for kiosks).	CSIRO Kick-Start provides $10k–$50k in matched funding for SMEs to conduct research with CSIRO . We will use this to engage CSIRO’s Data61 for advanced P2P software or their energy researchers for solar/storage optimization. For example, a Kick-Start project in 2025 could develop a custom low-bandwidth CRDT sync protocol tailored to our kiosks.
Mobile Network Hardening & Telecom Resilience Grants	Emergency Communications: Funding or support in kind for satellite links, backup	The federal government has committed $340M+ to improving communications resilience, via programs like

	communication equipment on kiosks to ensure connectivity during disasters.	the Mobile Network Hardening Program and Telecommunications Disaster Resilience Innovation grants . Rather than direct grants, these could materialize as partnerships with telcos or NBN Co. (e.g. expansion of the STAND satellite Wi-Fi to our kiosks). We will actively seek inclusion in such initiatives so that each kiosk might be outfitted with, say, an NBN SkyMuster satellite unit or funded to have battery-backed cell repeaters.
Redland City Council – Community Grants & Infrastructure Funds	Local Community Support: Small grants to support community engagement activities and minor capital works for kiosk sites.	Redland City offers grants up to $10k for community projects and also has a Capital Infrastructure Fund for community facilities . We will pursue these to fund community workshops (kiosk training sessions, etc.) and to co-fund site preparations (e.g. pouring a concrete pad, shading structure for a kiosk). Though modest in size, these grants demonstrate local buy-in and can cover “last mile” needs (like installing seating or signage at kiosk locations).
Federal and State Innovation & Inclusion Grants (2026)	Digital Inclusion & Skills: Broader programs aiming to bridge the digital divide or foster innovation in regions, which align with kiosk outcomes (digital access for	We will watch for calls like the Australian Digital Inclusion Program (if launched) or Queensland’s innovation challenges. For example, a Smart

	all, smart tech in a remote area).	Communities Challenge 2026 or Regional Digital Connectivity grants could fund enhancements like interactive content for the kiosks or additional connectivity (TV whitespace, etc.). Similarly, programs under the Medical Research Future Fund (MRFF) for digital health and the National Indigenous Australians Agency might support the Aura health integration or Indigenous data aspects . Each opportunity will be matched to the relevant kiosk feature – e.g. MRFF dementia care grants for the Aura kiosk trial, or a digital literacy grant to run training at kiosks for elders.

Table: Key funding opportunities and how they map to kiosk project needs. These include state programs (Advance Queensland), federal resilience and smart infrastructure grants, research partnerships, and local council support.

As shown, 2025–2026 is a ripe period to secure funding, with many programs aligning with our milestones:

- In the immediate term, Advance Queensland grants can kick-start development (Ignite Ideas) and support scale (Ignite+ Scale), leveraging the state's push to build an innovation economy in regions . The project's strong alignment with Indigenous empowerment and new tech makes it a "perfect fit" for Advance Qld's goals .

- For larger capital needs, the Disaster Ready Fund and Smart/Thriving Suburbs provide a pathway to millions in funding for exactly this kind of community resilience infrastructure . We will prepare a compelling case highlighting the kiosk network's innovation and co-benefits (e.g. economic, social, cultural) to stand out in these competitive grants. Notably, demonstrating the pilot kiosk and community buy-in (from earlier milestones) will strengthen these applications.

- By tapping CSIRO and research grants, we not only get funds but also validation and technical support. Engaging CSIRO via Kick-Start or even exploring an ARC Linkage project with universities (as the Pathways report suggests for R&D collaborations) could attract additional resources and cutting-edge expertise for the project's later

stages (like optimizing the digital twin algorithms for offline use).

- We'll also pursue non-grant funding in parallel: e.g. partnerships with tech companies for in-kind contributions (donation of solar panels or batteries), and social impact investors. The project's measurable social/environmental benefits (improved disaster outcomes, community capacity-building) make it attractive to impact investment – as noted in our strategy, a clear "Theory of Change" can unlock philanthropic capital . For instance, we might secure sponsorship from a foundation to fund kiosk units in exchange for data on how they improve community resilience.

A systematic, proactive grant strategy will be implemented – essentially making grant applications a "core operational activity" of the project team . Each grant will be targeted to the appropriate component: small local grants for community engagement, state innovation grants for tech development, and big federal grants for infrastructure rollout . This stacking of funding sources ensures no single source bears all the cost and that we meet matching requirements when they arise (for example, using council funds or CSIRO in-kind support to match federal dollars).

Importantly, we will time our applications to align with project readiness. For example, apply for DRF once the pilot is proven (so we have data to show its efficacy), and pursue Smart Suburbs funding once we have detailed plans and co-funding in hand. Early engagement with grant bodies is also planned – we'll reach out to program managers (e.g. at Advance Queensland, QRA, DESI) for feedback and to ensure our proposals tick the right boxes, since our strategy is to mirror public policy objectives (e.g. showing how kiosks deliver on Queensland's digital inclusion or disaster preparedness strategies) .

Through this blended funding approach, by 2026 we expect to have secured enough resources to implement the kiosks and sustain operations. The grants table will serve as a living document, updated as new opportunities (like any 2026 special resilience funds or tech challenges) emerge. Given the strong alignment – from local council support up to national innovation agendas – we are confident that the financial pathway for the kiosk network is robust.

Conclusion

North Stradbroke Island's offline-first kiosk network represents a pioneering model of community resilience – one that fuses ancient wisdom of self-reliance with cutting-edge decentralized technology. This comprehensive plan has outlined how a series of solar-powered "sovereign node" kiosks can keep Minjerribah connected, powered, and coordinated through any crisis. By fulfilling multiple roles (from emergency communication to democratic participation and resource sharing), the kiosks address both the acute shocks of disasters and the everyday needs of a digitally empowered island.

Crucially, this strategy is built on the strengths of the community and existing initiatives. It amplifies the Virtual Minjerribah digital twin by making it accessible and operational offline; it extends the Aura of Intelligence care system to those most vulnerable in disasters; it provides practical training and jobs through SETCo; and it honors Quandamooka

sovereignty by ensuring the community owns and controls the infrastructure and data. The plan's milestone framework ensures gradual, consultative implementation – starting small, proving value, and scaling up resilience in tandem with growing community capacity.

In an era of climate uncertainty and technological centralization, Minjerribah's approach can set a new benchmark: a small island showing the world how to weather storms (literal and metaphorical) through decentralization, local empowerment, and innovation. As one component of the island's broader vision of a "regenerative operating system" and "fully integrated, self-sustaining ecosystem" , the kiosk network will help safeguard not just infrastructure but the cultural and social fabric of the community. It is a tangible step toward the "Joyful, Resilient Abundance" that the Quandamooka people and fellow islanders are co-creating for future generations.

With community will, technical ingenuity, and the support of targeted grants and partnerships, North Stradbroke Island will become disaster-proof in a way that also enriches daily life. When the next cyclone roars or the next solar flare crackles, Straddie will stand as a shining example of resilience – its lights literally kept on by the sun, its people informed and united via kiosks that ask no permission to function, and its destiny firmly in its own hands. This is the promise of the Community Sovereign Kiosk plan: technology with Country at its heart, turning vulnerability into strength on Minjerribah.

Sources: The concepts and strategies in this plan draw on the Minjerribah Innovation Ecosystem documents and current governmental programs, including the Virtual Minjerribah simulation architecture , the Pathways to Government Engagement report , and funding programs like Advance Queensland and the Disaster Ready Fund . These sources and others have been cited throughout to provide evidence and alignment with established goals and successful precedents.